```
import seaborn as sns
import pandas as pd
import numpy as np
```

```
tips = sns.load_dataset("tips")

print("Population size:", len(tips))
tips.head()
```

```
Population size: 244
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

Next steps: Generate code with tips New interactive sheet

```
simple_sample = tips.sample(n=20, random_state=1)

print("Simple Random Sample")
print(simple_sample)
```

```
Simple Random Sample
     total_bill    tip     sex smoker   day    time  size
67         3.07   1.00  Female    Yes   Sat  Dinner     1
243       18.78   3.00  Female     No  Thur  Dinner     2
206       26.59   3.41    Male    Yes   Sat  Dinner     3
122       14.26   2.50    Male     No  Thur   Lunch     2
89        21.16   3.00    Male     No  Thur   Lunch     2
218        7.74   1.44    Male    Yes   Sat  Dinner     2
58        11.24   1.76    Male    Yes   Sat  Dinner     2
186       20.90   3.50  Female    Yes   Sun  Dinner     3
177       14.48   2.00    Male    Yes   Sun  Dinner     2
4         24.59   3.61  Female     No   Sun  Dinner     4
220       12.16   2.20    Male    Yes   Fri   Lunch     2
226       10.09   2.00  Female    Yes   Fri   Lunch     2
116       29.93   5.07    Male     No   Sun  Dinner     4
107       25.21   4.29    Male    Yes   Sat  Dinner     2
170       50.81  10.00    Male    Yes   Sat  Dinner     3
241       22.67   2.00    Male    Yes   Sat  Dinner     2
181       23.33   5.65    Male    Yes   Sun  Dinner     2
51        10.29   2.60  Female     No   Sun  Dinner     2
27        12.69   2.00    Male     No   Sat  Dinner     2
240       27.18   2.00  Female    Yes   Sat  Dinner     2
```

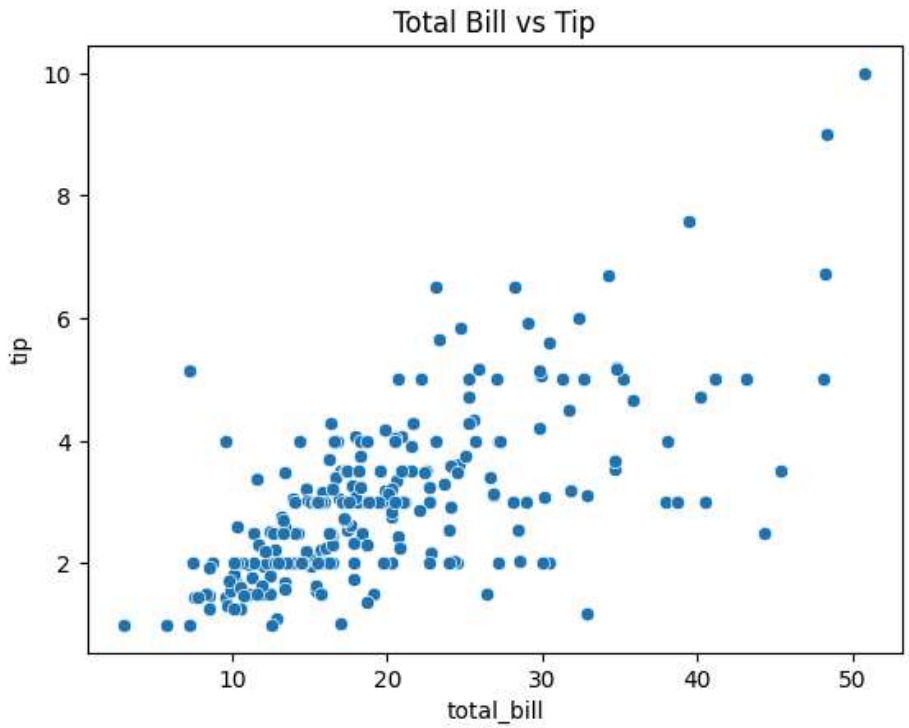
```
step = 5
systematic_sample = tips.iloc[::step]

print("Systematic Sample")
print(systematic_sample.head())
```

```
Systematic Sample
    total_bill   tip     sex smoker  day    time  size
0        16.99  1.01  Female     No  Sun  Dinner     2
5        25.29  4.71    Male     No  Sun  Dinner     4
10       10.27  1.71    Male     No  Sun  Dinner     2
15       21.58  3.92    Male     No  Sun  Dinner     2
20       17.92  4.08    Male     No  Sat  Dinner     2
```

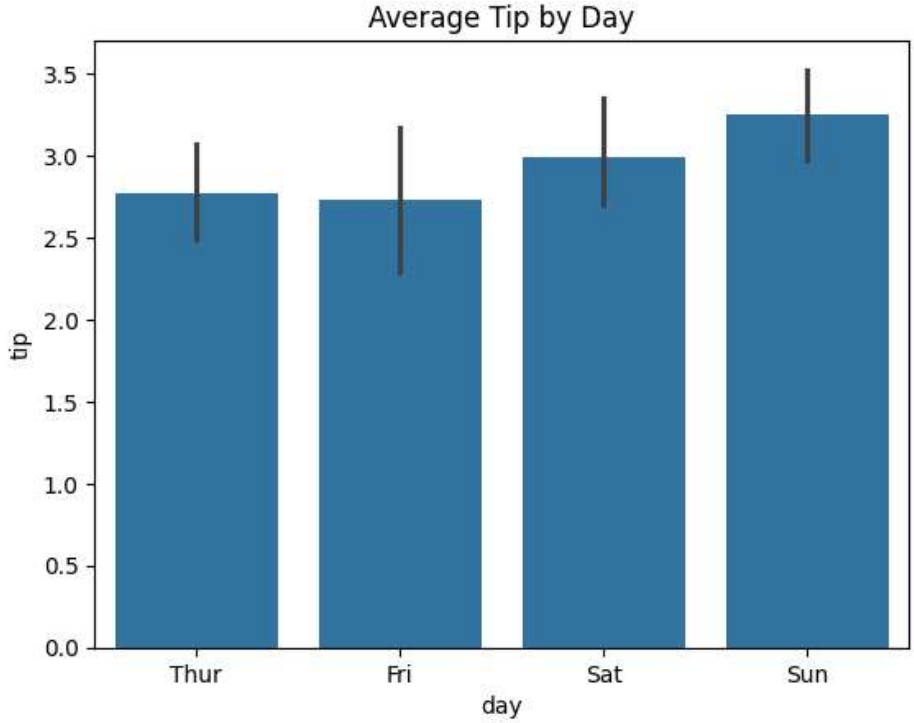
```
stratified_sample = tips.groupby("day").apply(lambda x: x.sample(5))

print("Stratified Sample")
print(stratified_sample)
```

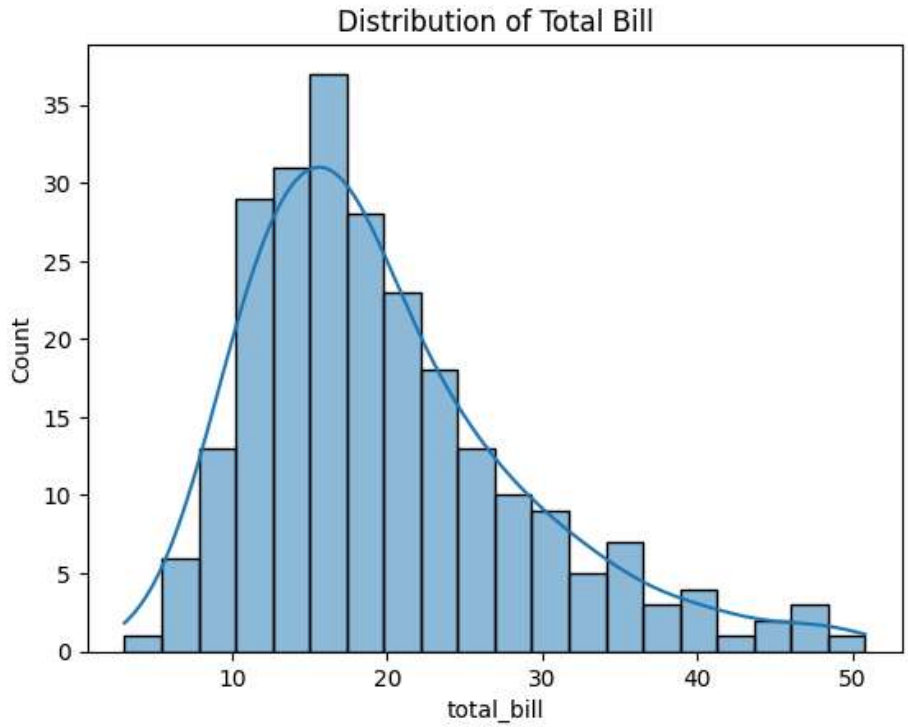
```
Stratified Sample
          total_bill   tip     sex smoker   day    time  size
day
Thur 120       11.69  2.31    Male     No  Thur   Lunch     2
     85        34.83  5.17  Female     No  Thur   Lunch     4
     135        8.51  1.25  Female     No  Thur   Lunch     2
     142       41.19  5.00    Male     No  Thur   Lunch     5
     145        8.35  1.50  Female     No  Thur   Lunch     2
Fri  90        28.97  3.00    Male    Yes   Fri  Dinner     2
     100       11.35  2.50  Female    Yes   Fri  Dinner     2
     95        40.17  4.73    Male    Yes   Fri  Dinner     4
     92         5.75  1.00  Female    Yes   Fri  Dinner     2
     220       12.16  2.20    Male    Yes   Fri   Lunch     2
Sat  20        17.92  4.08    Male     No   Sat  Dinner     2
     234       15.53  3.00    Male    Yes   Sat  Dinner     2
     213       13.27  2.50  Female    Yes   Sat  Dinner     2
     104       20.92  4.08  Female     No   Sat  Dinner     2
     21        20.29  2.75  Female     No   Sat  Dinner     2
Sun  43         9.68  1.32    Male     No   Sun  Dinner     2
     46        22.23  5.00    Male     No   Sun  Dinner     2
     0         16.99  1.01  Female     No   Sun  Dinner     2
     9         14.78  3.23    Male     No   Sun  Dinner     2
     112       38.07  4.00    Male     No   Sun  Dinner     3
/tmp/ipykernel_381/1084610073.py:1: FutureWarning: The default of observed=False is deprecated and
  stratified_sample = tips.groupby("day").apply(lambda x: x.sample(5))
/tmp/ipykernel_381/1084610073.py:1: DeprecationWarning: DataFrameGroupBy.apply operated on the gro
  stratified_sample = tips.groupby("day").apply(lambda x: x.sample(5))
```

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.scatterplot(data=tips, x="total_bill", y="tip")

plt.title("Total Bill vs Tip")
plt.show()
```



```
sns.barplot(data=tips, x="day", y="tip")

plt.title("Average Tip by Day")
plt.show()
```




```
sns.histplot(tips["total_bill"], bins=20, kde=True)

plt.title("Distribution of Total Bill")
plt.show()
```